

A Comprehensive Break of the Tropical Matrix-Based Signature Scheme

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March 15, 2026

Abstract

In the recent preprint [GMS26], Grigoriev, Monico and Shpilrain proposed a digital signature protocol based on matrices over the tropical integer semiring. We show that the scheme can be completely broken. We present a chosen-hash forgery, a malleability attack that breaks strong unforgeability, and an SMT-based key recovery attack demonstrating that a single valid signature together with the public key suffices to recover a full private key in within one second for the recommended parameters. Our attacks exploit algebraic properties of the signing equations rather than the claimed NP-hard problem. Independent work by Sferlazza [1] concurrently identified related vulnerabilities.

Keywords: digital signature, tropical semiring, cryptanalysis, matrix factorization, malleability, SMT solving

1 Introduction

The search for post-quantum cryptographic primitives has led researchers to explore exotic algebraic structures as potential platforms. Among these, tropical semirings—where addition is replaced by the minimum operation

and multiplication by ordinary addition—have attracted attention due to their simplicity and the potential for highly efficient implementations [2]. In a recent preprint, Grigoriev, Monico, and Shpilrain [3] proposed a digital signature scheme built upon matrices over the tropical integer semiring. The authors claim that the security of their construction rests on the NP-complete problem of factoring a tropical matrix into a product of two matrices of given dimensions [4].

In this paper, we perform a comprehensive security analysis of the GMS26 digital signature scheme. We demonstrate that the scheme fails to achieve standard security notions—including existential unforgeability (EUF-CMA), strong unforgeability (SUF-CMA), and key confidentiality—through a series of increasingly powerful attacks. Our contributions are threefold:

- **Forgery attacks:** We present an existential forgery in the chosen-hash model (Theorem 4.1) that solves a tropical linear system in polynomial time. Furthermore, we show that the scheme is fundamentally malleable (Theorem 4.2), allowing an adversary to generate arbitrarily many distinct valid signatures for any observed message, thereby breaking strong unforgeability.
- **SMT-based key recovery:** We demonstrate that a *single* valid signature, together with the public key, suffices to recover a full private key. Using an SMT solver (Z3), we recover a valid key in under one second for the recommended parameters $(m, n, k) = (8, 8, 7)$ (Section 5). The public key equation $T = X \otimes Y$ is essential for this attack; without it, the recovered key does not match the public key and cannot be used for universal forgery. Once a key satisfying $T = X \otimes Y$ is recovered, universal forgeries indistinguishable from honest signatures can be generated for any message.
- **Structural weaknesses:** We identify inherent design flaws—linearity of the verification equation, idempotence of $\oplus$, and probabilistic masking—that make the above attacks possible. These flaws are independent of the claimed NP-hard problem and cannot be repaired by simple parameter tweaks.

In concurrent and independent work, Sferlazza [1] also identified vulnerabilities in the GMS26 scheme, including a standard-model forgery and an SMT-based key recovery attack using one signature together with the public key. Previous attacks on tropical signature schemes, such as the one

by Brown and Monico [5] against an earlier construction, also exploited algebraic weaknesses. Our work complements and extends these results: we introduce the first malleability attack (breaking SUF-CMA), provide experimental confirmation that the public key is necessary for unique key recovery, and offer a more detailed analysis of the scheme’s structural deficiencies. An earlier version of this paper contained a flawed probabilistic analysis; we have removed it entirely and now refer the reader to Sferlazza’s correct treatment [1] for probabilistic leakage.

The remainder of this paper is organised as follows. Section 2 recalls the necessary background on tropical algebras and provides a self-contained description of the GMS26 signature scheme. Section 3 identifies fundamental design flaws inherent to the construction. Section 4 presents our forgery attacks, including the chosen-hash existential forgery and the malleability attack. Section 5 describes the SMT-based exact key recovery attack and presents experimental results. Section 6 provides a detailed comparison with prior work, and Section 7 concludes.

2 Preliminaries

2.1 Tropical Semiring and Matrices

Let $\mathbb{Z}_{\min,+} = (\mathbb{Z} \cup \{\infty\}, \oplus, \otimes)$ denote the **tropical integer semiring**, where the operations are defined for all $x, y \in \mathbb{Z} \cup \{\infty\}$ as

$$x \oplus y = \min\{x, y\}, \quad x \otimes y = x + y.$$

The additive identity (zero element) is $\epsilon = \infty$, and the multiplicative identity is 0. These operations extend naturally to matrices. For matrices A, B of identical dimensions, $(A \oplus B)_{ij} = A_{ij} \oplus B_{ij}$. For matrices $A \in \mathbb{Z}^{m \times p}$ and $B \in \mathbb{Z}^{p \times n}$, their **tropical product** is defined entrywise as

$$(A \otimes B)_{ij} = \bigoplus_{k=1}^p (A_{ik} \otimes B_{kj}) = \min_{1 \leq k \leq p} \{A_{ik} + B_{kj}\}.$$

Following [3], we denote by $\mathcal{M}_{r,s}^{\leq N}$ the set of $r \times s$ matrices with integer entries in the range $[0, N]$.

2.2 Description of the GMS26 Signature Scheme

We now recall the GMS26 digital signature scheme with the recommended parameters $(m, n, k) = (8, 8, 7)$. We adopt the notation of [3] throughout for consistency and refer the reader to the original paper for further details.

Key generation. The signer selects two matrices uniformly at random:

$$X \xleftarrow{\$} \mathcal{M}_{m,k}^{\leq 255}, \quad Y \xleftarrow{\$} \mathcal{M}_{k,n}^{\leq 255}.$$

The **private key** is $\mathbf{sk} = (X, Y)$. The **public key** is the tropical product

$$\mathbf{pk} = T = X \otimes Y \in \mathcal{M}_{m,n}^{\leq 510}.$$

Signing algorithm. Given a message m with timestamp t , the signer computes:

1. The hash matrix $M = \mathcal{H}(m||t) \in \mathcal{M}_{n,m}^{\leq 255}$, where $\mathcal{H}$ is a cryptographic hash function (e.g., SHA3-512) with a deterministic procedure to convert its output to an 8×8 matrix of integers in $[0, 255]$.
2. Ephemeral secret matrices sampled uniformly:

$$U \xleftarrow{\$} \mathcal{M}_{n,k}^{\leq 255}, \quad V \xleftarrow{\$} \mathcal{M}_{k,m}^{\leq 255},$$

resampling until the following conditions hold:

$$(M \otimes X) \oplus U \neq U, \quad (Y \otimes M) \oplus V \neq V.$$

3. The signature components:

$$\begin{aligned} A &= (M \otimes X) \oplus U, & B &= (Y \otimes M) \oplus V, \\ P &= X \otimes V, & R &= U \otimes Y, \\ S &= U \otimes V. \end{aligned}$$

4. The signature is the tuple $\sigma = (M, A, B, P, R, S)$.

Verification algorithm. Given a public key T , a message m with timestamp t , and a candidate signature $\sigma = (M, A, B, P, R, S)$, the verifier:

1. Checks that $M = \mathcal{H}(m||t)$. If not, reject.
2. Computes $W = (M \otimes T \otimes M) \oplus (M \otimes P) \oplus (R \otimes M) \oplus S$.
3. Accepts the signature if and only if the following two conditions hold:

$$A \otimes B = W, \quad (\text{V1})$$

$$A \otimes B \neq S. \quad (\text{V2})$$

The scheme’s efficiency derives from the fact that tropical multiplication is ordinary addition, avoiding the computational cost of integer multiplication. The authors claim that security rests on the NP-hardness of factoring the public key T into a product of an $m \times k$ matrix and a $k \times n$ matrix [4]. A reference implementation of the scheme, including the parameter choices and matrix operations, is available at [6].

3 Design Flaws

The GMS26 signature scheme exhibits several structural weaknesses that are independent of the underlying tropical matrix factorization problem. These flaws are exploited by the attacks presented in this paper and highlight the gap between the claimed hardness assumption and the concrete security of the protocol. Moreover, they arise not from poor parameter choices but from inherent limitations of the min-plus semiring itself.

Linearity of the verification equation. The verification condition

$$A \otimes B = (M \otimes T \otimes M) \oplus (M \otimes P) \oplus (R \otimes M) \oplus S$$

is linear in the tropical sense: each term on the right-hand side is a tropical product involving only public information (M, T) and the signature components (P, R, S) . The left-hand side $A \otimes B$ is also a product of two matrices provided in the signature. Because tropical multiplication distributes over addition, every component contributes independently via a separate monomial. This linear structure allows an adversary to manipulate the signature components independently, as demonstrated in the malleability attack (Theorem 4.2) and the chosen-hash forgery (Theorem 4.1).

Insufficient rejection sampling. The signing algorithm requires that $(M \otimes X) \oplus U \neq U$ and $(Y \otimes M) \oplus V \neq V$, i.e., that the masks U, V are not dominating the entire matrices. However, as observed by Sferlazza [1], this condition does not prevent the pathological case where $A \otimes B = U \otimes V$ even though $A \neq U$ and $B \neq V$. Such signatures are valid according to the signing procedure but are rejected by the verification condition $A \otimes B \neq S$. This inconsistency means that honest signatures can be rejected with non-negligible probability (experimentally up to 9% for some parameter sets), violating the basic correctness requirement of a signature scheme.

Masking is probabilistic and leaks information. The masks U and V are intended to hide the private key X and Y by being combined via the tropical $\oplus$ (min) operation. However, because $\oplus$ selects the minimum, a mask entry hides the corresponding key-derived value only when it is strictly smaller. In all other cases, the value $A_{i,j}$ equals $(M \otimes X)_{i,j}$, directly revealing linear combinations of X . This probabilistic leakage is unavoidable: over multiple signatures the probability that a given entry remains hidden decays exponentially, and the entire key eventually becomes exposed. The same reasoning applies to Y via the B matrices. A rigorous analysis of this leakage is given in Sferlazza [1].

Idempotence of $\oplus$ enables malleability. The tropical addition $\oplus$ is idempotent ($x \oplus x = x$) and not cancellative. This property is central to the malleability attack: because the verification equation is a minimum of several terms, increasing one term that is not the unique minimum does not change the result. An adversary can therefore modify a signature component without invalidating the equation, producing arbitrarily many distinct valid signatures for the same message. This directly breaks strong unforgeability (SUF-CMA).

Structural impossibility. These defects are not accidental; they reflect fundamental limitations of any signature scheme whose verification equation is a tropical polynomial of bounded degree. In such a setting, linearity enables component-wise manipulation, idempotence prevents detection of certain modifications, and the non-invertibility of min precludes effective masking. Any attempt to repair the protocol must address these algebraic roots rather than simply increasing matrix dimensions.

4 Forgery Attacks

In this section, we present two distinct forgery attacks against the GMS26 signature scheme. First, we demonstrate an existential forgery in the chosen-hash model, where the adversary can influence the output of the hash function. Second, we show that the scheme is malleable, allowing an adversary to generate arbitrarily many distinct valid signatures for a previously signed message—thereby breaking strong unforgeability (SUF-CMA).

4.1 Chosen-Hash Existential Forgery

We first consider the chosen-hash model, in which the adversary is allowed to influence the hash output $M = \mathcal{H}(m||t)$. This model is standard for exposing structural weaknesses in signature schemes [7].

Theorem 4.1 (Chosen-hash existential forgery). *In the GMS26 signature scheme, suppose an adversary can choose the matrix $M \in \mathcal{M}_{n,m}^{\leq 255}$ representing the hash of a message. Then there exists a polynomial-time algorithm that produces a valid signature on M without knowledge of the private key (X, Y) .*

Proof. Let $T \in \mathcal{M}_{m,n}^{\leq 510}$ be the public key. The adversary proceeds as follows:

1. Pick arbitrary matrices $\tilde{X} \in \mathcal{M}_{m,k}^{\leq 255}$ and $\tilde{Y} \in \mathcal{M}_{k,n}^{\leq 255}$.
2. Solve for $M \in \mathcal{M}_{n,m}^{\leq 255}$ satisfying the tropical linear system

$$(\tilde{X} \otimes \tilde{Y}) \otimes M = T \otimes M.$$

This is a tropical linear system that can be solved in polynomial time using, for example, the methods described in [8].

3. Choose arbitrary ephemeral matrices $U \in \mathcal{M}_{n,k}^{\leq 255}$ and $V \in \mathcal{M}_{k,m}^{\leq 255}$.
4. Construct the forged signature components:

$$\begin{aligned} A &= (M \otimes \tilde{X}) \oplus U, & B &= (\tilde{Y} \otimes M) \oplus V, \\ P &= \tilde{X} \otimes V, & R &= U \otimes \tilde{Y}, \\ S &= U \otimes V. \end{aligned}$$

5. Output the forged signature $\sigma = (M, A, B, P, R, S)$.

We now verify that σ satisfies the verification conditions. By construction,

$$A \otimes B = (M \otimes \tilde{X} \otimes \tilde{Y} \otimes M) \oplus (M \otimes P) \oplus (R \otimes M) \oplus S.$$

Since $\tilde{X} \otimes \tilde{Y}$ was chosen arbitrarily, condition (V1) of the GMS26 scheme requires that $A \otimes B$ equal

$$W = (M \otimes T \otimes M) \oplus (M \otimes P) \oplus (R \otimes M) \oplus S.$$

Our choice of M ensures that $M \otimes \tilde{X} \otimes \tilde{Y} \otimes M = M \otimes T \otimes M$, hence $A \otimes B = W$. Condition (V2) requires $A \otimes B \neq S$. Since U and V are

arbitrary, the probability that $A \otimes B = S$ is negligible for the recommended parameter sizes; if it occurs, the adversary can simply resample U and V . Thus, the forgery is valid with high probability. $\square$

This attack demonstrates that the GMS26 scheme does not achieve existential unforgeability under chosen-message attacks (EUF-CMA) if the adversary has even limited influence over the hash output.

Remark. The existence of a solution M to the tropical linear system $(\tilde{X} \otimes \tilde{Y}) \otimes M = T \otimes M$ is not guaranteed for arbitrary choices of $\tilde{X}, \tilde{Y}$. However, in practice one can sample $\tilde{X}, \tilde{Y}$ randomly until a solution exists; for the small parameter sizes considered here this succeeds quickly. Moreover, the scheme is already broken by the attacks in Sections 4.2 and 5, so this limitation does not affect the overall conclusion.

4.2 Malleability: Breaking Strong Unforgeability

Strong unforgeability (SUF-CMA) requires that an adversary cannot produce a *new* valid signature for a message that has already been signed. We show that the GMS26 scheme is malleable and therefore fails to satisfy this notion.

Theorem 4.2 (Malleability). *Let $\sigma = (M, A, B, P, R, S)$ be a valid signature on a message represented by M in the GMS26 scheme. Then an adversary without knowledge of the private key can construct arbitrarily many distinct valid signatures for the same message M .*

Proof. Since σ is valid, it satisfies verification condition (V1) from [3]:

$$C := A \otimes B = (M \otimes T \otimes M) \oplus (M \otimes P) \oplus (R \otimes M) \oplus S,$$

where T is the public key. For each entry (i, j) , the value $C[i, j]$ is the minimum of four numbers:

$$C[i, j] = \min\{(M \otimes T \otimes M)[i, j], (M \otimes P)[i, j], (R \otimes M)[i, j], S[i, j]\}.$$

Except in degenerate cases, at least one term other than $S[i, j]$ attains this minimum. Choose an index (i_0, j_0) where $S[i_0, j_0] > C[i_0, j_0]$ and another term achieves the minimum. Define a modified signature $\sigma' = (M, A, B, P, R, S')$ where

$$S'[i, j] = \begin{cases} S[i, j] & \text{if } (i, j) \neq (i_0, j_0), \\ C[i_0, j_0] + \delta & \text{if } (i, j) = (i_0, j_0), \end{cases}$$

for any integer $\delta > 0$ such that the new value remains within the allowed range $[0, 510]$ (as specified in [3]). At (i_0, j_0) we have

$$\min\{(M \otimes T \otimes M)[i_0, j_0], (M \otimes P)[i_0, j_0], (R \otimes M)[i_0, j_0], C[i_0, j_0] + \delta\} = C[i_0, j_0],$$

since $C[i_0, j_0] + \delta > C[i_0, j_0]$ and another term still attains $C[i_0, j_0]$. All other entries are unchanged, so condition (V1) holds for σ' .

Condition (V2) requires $A \otimes B \neq S'$. By construction, $(A \otimes B)[i_0, j_0] = C[i_0, j_0] \neq C[i_0, j_0] + \delta = S'[i_0, j_0]$, so the inequality holds. Since δ can be chosen arbitrarily (e.g., $\delta = 1, 2, \dots, 510 - C[i_0, j_0]$), this yields arbitrarily many distinct valid signatures for the same message M . $\square$

This malleability attack exploits the idempotence of the tropical addition operation $\oplus$ and the linear structure of the verification equation. It demonstrates that the GMS26 scheme fails to achieve strong unforgeability, a standard security requirement for digital signatures.

5 Exact Key Recovery via SMT Solving

We now present a practical attack that, given only one valid signature and the public key, recovers a full private key using an SMT solver. The attack works for the recommended parameters $(m, n, k) = (8, 8, 7)$ and does not rely on any assumptions about the randomness of masks. The system includes the public key equation $T = X \otimes Y$, which makes the constraints sufficiently overdetermined to uniquely determine a private key that enables universal forgeries.

5.1 System of Equations from a Single Signature

Let $\sigma = (M, A, B, P, R, S)$ be a valid signature on a message with hash matrix M . From the definition of the GMS26 scheme [3] there exist unknown matrices

$$X \in \mathcal{M}_{m,k}^{\leq 255}, \quad Y \in \mathcal{M}_{k,n}^{\leq 255}, \quad U \in \mathcal{M}_{n,k}^{\leq 255}, \quad V \in \mathcal{M}_{k,m}^{\leq 255}$$

satisfying the following tropical equations:

$$A = (M \otimes X) \oplus U, \tag{1}$$

$$B = (Y \otimes M) \oplus V, \tag{2}$$

$$P = X \otimes V, \tag{3}$$

$$R = U \otimes Y, \tag{4}$$

$$S = U \otimes V, \tag{5}$$

together with the public key relation

$$T = X \otimes Y. \quad (6)$$

All entries are integers in $[0, 255]$ for X, Y, U, V and in $[0, 510]$ for the derived matrices. This yields a finite system of tropical polynomial equations: 368 equations in 224 unknowns, which is heavily overdetermined. Because the system corresponds to an honestly generated signature, it is guaranteed to have at least one solution — namely the original private key and masks. Any solution (X', Y') satisfying $T = X' \otimes Y'$ enables universal forgery (see §5.6).

5.2 SMT Encoding

We encode each tropical equality $c = \min(a_1, \dots, a_t)$ as a conjunction of linear inequalities together with a clause enforcing that c equals one of the a_i :

$$c \leq a_i \quad \forall i, \quad \text{and} \quad \bigvee_{i=1}^t (c = a_i).$$

All variables are declared as integers with bounds $[0, 255]$. This encoding is standard in SMT-based cryptanalysis [9]. For the recommended parameters, the resulting instances contain approximately 1500 Boolean constraints and are solved efficiently by modern SMT solvers.

5.3 Why a Single Signature Suffices for Key Recovery

We now provide a theoretical justification for why a single valid signature together with the public key yields a solvable SMT system.

Theorem 5.1 (Constraint Determination from a Single Signature). *Let (T, σ) be the public key and a valid signature in the GMS26 signature scheme with parameters $(m, n, k) = (8, 8, 7)$, where $\sigma = (M, A, B, P, R, S)$. Then the matrices X, Y, U, V satisfying equations (1)–(6) form a system of tropical polynomial constraints that is heavily overdetermined. In particular, the number of independent constraints exceeds the number of unknown variables, and the system admits a solution corresponding to a valid private key (X, Y) . Consequently, a satisfiability modulo theories (SMT) solver can recover such a key in practice for the recommended parameters.*

Proof. The unknown variables are the entries of X (56 variables), Y (56), U (56) and V (56), totalling 224 unknown integers bounded in $[0, 255]$. A

single signature provides the following matrix equations:

$$\begin{aligned}
A &= (M \otimes X) \oplus U, & |A| &= 56, \\
B &= (Y \otimes M) \oplus V, & |B| &= 56, \\
P &= X \otimes V, & |P| &= 64, \\
R &= U \otimes Y, & |R| &= 64, \\
S &= U \otimes V, & |S| &= 64, \\
T &= X \otimes Y, & |T| &= 64.
\end{aligned}$$

Counting entries gives 368 scalar constraints. Each tropical equality expands into a set of linear inequalities together with a disjunction enforcing equality with one of the candidate minima. After encoding these constraints in SMT, the resulting instance is a finite system of integer inequalities with bounded domains.

Because the system corresponds to a valid signature generated by the scheme, it is guaranteed to have at least one solution – namely the original private key (X, Y) together with the masks (U, V) used during signing. Modern SMT solvers can efficiently search the bounded domain and recover such a solution. Therefore, given a single signature and the public key, an SMT solver can recover a private key consistent with the signature. $\square$

Remark 5.2. Although the system may admit multiple solutions due to the tropical structure of the equations, any recovered pair (X', Y') satisfying $T = X' \otimes Y'$ enables universal forgery of signatures under the public key T (see §5.6). Hence recovering any such solution is sufficient to break the scheme.

5.4 Experimental Results

We implemented the encoding using the Z3 SMT solver (version 4.12.2). Experiments were run on Google Colab (Intel Xeon CPU @ 2.20GHz). For each trial we generated a random key pair and a random signature, and attempted to recover a valid private key using the full system including the public key equation. Table 1 summarises the results over 100 independent trials.

The attack achieves 100% success even with a single signature, recovering a private key that satisfies $T = X \otimes Y$ and therefore enables universal forgery. The runtime grows approximately linearly with the number of signatures, as expected when adding more variables and constraints.

Table 1: SMT-based key recovery (with public key): success rate and average runtime over 100 trials.

N (signatures)	Success rate	Avg. time (s)
1	100%	0.81
2	100%	1.39
3	100%	2.10
4	100%	2.90

5.5 Comparison with Prior Work

Sferlazza [1] independently proposed an SMT-based key recovery attack that also uses a single signature together with the public key. His experiments reported that the solver almost always finishes within one minute. Our results confirm and strengthen this finding: we observe 100% success in under one second. The small discrepancy in runtime can be attributed to differences in encoding details and solver versions; the essential fact remains that the GMS26 scheme is completely broken by a single observed signature together with the public key.

5.6 Universal Forgery after Key Recovery

Once a valid private key (X', Y') is recovered with $X' \otimes Y' = T$, forging a signature on any message m^* (with hash $M^* = \mathcal{H}(m^* || t^*)$) is trivial:

1. Choose arbitrary ephemeral masks $U^* \in \mathcal{M}_{n,k}^{\leq 255}$ and $V^* \in \mathcal{M}_{k,m}^{\leq 255}$.
2. Compute

$$\begin{aligned}
 A^* &= (M^* \otimes X') \oplus U^*, & B^* &= (Y' \otimes M^*) \oplus V^*, \\
 P^* &= X' \otimes V^*, & R^* &= U^* \otimes Y', \\
 S^* &= U^* \otimes V^*.
 \end{aligned}$$

3. Output $\sigma^* = (M^*, A^*, B^*, P^*, R^*, S^*)$.

This signature satisfies all verification equations and is statistically indistinguishable from an honest signature, because the distribution of (U^*, V^*) matches that of the legitimate signing algorithm.

6 Comparison with Prior Work

Our attacks complement and extend the recent cryptanalysis by Sferlazza [1], who independently identified a standard-model forgery and an SMT-based

key recovery attack using one signature together with the public key. Table 2 summarises the differences and commonalities between the two works.

Table 2: Comparison of cryptanalytic results against the GMS26 signature scheme.

Attack Vector	This Work	Sferlazza [1]
Forgery attacks		
Chosen-hash model forgery	✓ (Theorem 4.1)	—
Standard model forgery	—	✓
Malleability (SUF-CMA failure)	✓ (Theorem 4.2)	—
Key recovery (probabilistic)		
From multiple signatures	—	✓ (Section 4.2 of [1])
Key recovery (SMT-based)		
From 1 signature (with public key)	✓ (Section 5)	✓ (Section 4.1 of [1])
From 2 or more signatures	✓ (Section 5)	✓ (implied)
After key recovery		
Universal forgery capability	✓ (Section 5.6)	✓

As the table shows, both works demonstrate that the GMS26 scheme is completely broken. Sferlazza’s probabilistic analysis (Section 4.2 of [1]) is mathematically sound and experimentally validated; we therefore removed our own flawed probabilistic treatment from the present version. Our main novel contributions are the chosen-hash forgery (Theorem 4.1) and, most notably, the malleability attack (Theorem 4.2) which breaks strong unforgeability—a property not considered in [1].

Regarding SMT-based key recovery, both papers confirm that a single signature together with the public key suffices to recover a valid private key. We also investigated the possibility of key recovery without the public key (the NOPK variant). While the solver always finds a solution to the signature equations, it does not yield a key that matches the public key $T = X \otimes Y$, and therefore cannot be used for universal forgery. This highlights the essential role of the public key in uniquely determining the private key and underscores that an attacker must have access to the public key to perform a meaningful break.

In summary, the two works are complementary and together provide a complete picture of the scheme’s insecurity.

7 Conclusion

We have demonstrated that the GMS26 tropical signature scheme is completely broken. A single observed signature, together with the public key, suffices to recover a full private key via SMT solving in under one second, enabling universal forgeries (Section 5). The scheme is also malleable, breaking strong unforgeability (Theorem 4.2), and vulnerable to a chosen-hash forgery (Theorem 4.1). These attacks exploit inherent algebraic weaknesses rather than the claimed NP-hard problem. Our findings complement independent work by Sferlazza [1], who identified similar vulnerabilities. We have removed a flawed probabilistic analysis from an earlier version and refer the reader to [1] for a correct treatment.

Our implementation is available at <https://github.com/sjchavhanmath17-coder/tropical-signature-cryptanalysis>.

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